[Lecture Note] Entering and Organizing Quantitative Data

GE2021 Research Technology, Spring 2025

Chad (Chungil Chae)

Fri, 21 March 2025

Learning Objectives

- 9-1 Explain the purpose of entering and organizing data logically.
- 9-2 Summarize the steps needed to prepare to enter data.
- 9-3 Describe how to organize and input variables and their information.

SHORT VIDEO INTRODUCTION

https://www.youtube.com/watch?v=hQ5rSzTP1uY

The Purpose of Entering and Organizing Quantitative Data

- There are two principles involved in organizing data:
  - Place the results in various categories where they belong
  - Remove or fix any information that has the potential to intrude into the study or analysis.
- The first step in organizing data is to enter them in a readable and meaningful form.

Preparing for Data Entry

- The task of entering data. Can be repetitive and time-consuming, but it is crucial, and even a small mistake can impact the outcome.

- Logical Formatting
  - Take a critical look at the survey questions before beginning data entry to save time and frustration.
  - Determine how the variables can best be organized to ensure the data analysis phase is simple and straightforward.

Software Packages

- Data will be organized into a spreadsheet, and researchers often enter their data directly into their software of choice to avoid file corruption.
- Common programs for entering and analyzing data are Excel, Statistical Pack for Social Science (SPSS), Statistical Analysis System (SAS), Stata, R, Minitab, JMP, and more.
- Some software programs are better for analyzing rather than entering data; it is wise to examine the options before starting.
  - Different pieces of software have different strengths and weaknesses in data analysis.
  - It may be useful to do some internet research and determine which type of spreadsheet, like Excel, is the most easily imported into the analysis software of choice.

Unique Identification

- Assign a unique identification number to every participant or case in the study.
- Each complete questionnaire requires an identification number, which should be listed in the first column of the spreadsheet.
  - Each column represents a variable, and the first variable must be the identification numbers.
  - Every row will represent one participant or one case.

Missing Data

- It is common for people to leave at least one question of a questionnaire unanswered.
- Missing information can be important for analysis because it can signal an underlying problem with a specific question.
- When recording missing information, one possibility is to leave the variable blank, or another is to put an eye-catching, specific number that is not included in the standardized variable code.

Organizing and Inputting Variables

Coding and Codebooks

- A number of other minor decisions need to be made before data entry can begin.
- First, the researcher will need to determine how the responses to specific questions will be represented on the spreadsheet.
- By coding the information, the researcher can simplify raw information in a way that is easier to analyze.
- Nominable variable: also called qualitative variable; variables where answers are not mathematically related to each other.
  - Nominal variables are one of two subtypes of categorical variables.
  - Examples: gender, race, religion, and so on.
- Categorical variables: variables with answers that can be organized around various, often limited, categories.
- Ordinal variable: the other type of categorical variable; responses do not have a numeric value, but instead represent an ordered sequence of responses.
  - Example: Strongly agree, somewhat agree, somewhat disagree, strongly disagree.
- Numerical variable: also known as quantitative variables; expressed with numerical values in the original data
  - Examples: age, income, height, and so on.
  - The dataset must be consistent for analyzing information, and determining how to represent every single variable in a numeric fashion is a fundamental part of coding.
  - There are two types of numerical variables:
    - Interval: cannot express a ratio between numbers because of the lack of a true zero.
      - Example: Temperature
    - Ratio: has a true zero point.
      - Example: number of children a person has.
- Codebook: the key to the codes in the study and how they correlate with participant responses.
  - Keep in a separate, secure file.
  - Include as much detail as possible about the coding process, such as the original coding decisions and the reasoning behind them.

Variable Names

- Each variable needs to be identified because it can be easily confused with the variable next to it.
- Best practices for naming variables in a dataset to ensure the spreadsheet is compatible with various analytical software packages:
  - Names should typically be letters from the alphabet, avoiding special symbols or characters.
  - The variable name should be a single word, but may represent a combination of words.
    - Example: “raceethnic” for race and ethnicity

- Numbers may be a part of the variable name, but ideally they should be incorporated into a single word.
  - Example: “employm1” for primary employment position
  - The length of the variable should be short; maximum of 30-40 characters.

Variable Descriptions

- Data entry in SPSS or SAS will have boxes designated for short descriptions of variables, but if using Excel, the researcher may want to use an additional Word document to log descriptive records.
- The description should include information on how the variable was measured, a list of response choices, and other significant notes.

Notations for Responses

- Before data entry begins, the researcher needs to identify the notation for each answer choice.
- Many statistical software packages have a place to store information about answer choices, but if using Excel, it is a good idea to store notations separately.
- The notations reminds the researcher at any given moment what the numbers in the variable column stand for; however, it is not a substitute for the codebook.

Scales

- Psychometric scales are a universally accepted measure that ensures accuracy and avoids subjectivity.
- All scales should be consistent, even if the concepts are unrelated.
  - Example: making sure all question answers go from “strongly agree” to “strongly disagree” instead of the opposite order.
- Make sure to repeatedly save and back up data.

Class Activities:

- In this activity, you will be gathering and organizing data collected from students in your class. Use the table below to collect this information (ask students to handwrite their answers into one table that you circulate among all the students) and add additional rows to accommodate your class size. Photocopy the completed table and circulate it to all your students. Ask each student to enter the data exactly as collected from the table into a program like Excel. Ask the students to prepare the data for analysis by creating appropriate variable names for each variable. In order to anonymize the data, ask them to generate a unique ID for each student. As a class, discuss the data that was obtained from this exercise.
  - Do you notice any problems with your data?

- Did you encounter any challenges with data entry?
- Student name What is your height?
- What is your age? What is your eye color?
- What is your hair color?
- What is your favorite ice cream flavor?
- Where were you born?

Student name	What is your height?	What is your age?	What is your eye color?	What is your hair color?	What is your favorite ice cream flavor?	Where were you born?

- Identify each variable type from the table above. For example, is height a nominal, ordinal, interval, or ratio variable? Do the same for all the variables.

- Reflect on this exercise. What would you do differently if you were to repeat this exercise?

House Keeping

Announcement

Assignment

Assignment1: Assignment Title

- **What to do**

  - A
  - B

- **Format and Requirement**
  - PDF format, < ?? pages
  - file name should be include your student id and name
    - stuID_name_title.pdf (e.g. 1111111_ChungilChae_SelfIntroduction.pdf)
  - APA, Double-Space, No title page, 12 points, Reference section if necessary
  - Chinese name in English, English name, Student ID, Class name and section (e.g. GE2021, W09; MGS3001, W01)

- due date
  - by Date(DAY) 11:59PM
  - NO LATE SUBMISSION ALLOWED!!!!

Reference